Determining the effect of pH on Cellobiases reaction rate for Biofuel production

Aim:

To determine the effect of pH on the reaction rate of the cellobiase enzyme for catalysis of cellulose into glucose.

Introduction

Biofuels, such as bioethanol, biogas, or biodiesel, are fuels produced from renewable biomass. In order for microorganisms to convert biomass into biofuels through fermentation, certain biomass components, most prominently cellulose and hemicellulose, have to be broken down into smaller, fermentable constituents, most importantly glucose. They achieve this with a unique class of enzymes called cellulases.

Enzymes are biological, protein-made catalysts which help living systems speed up chemical reactions. Cellobiases (or β–glucosidases) belong to cellulase enzyme family and catalyse final break down of cellulose into the fermentable glucose molecules. They catalyse the cleavage of the β–(1,4') glycosidic bond of cellobiose (substrate) into two molecules of glucose. They are actively being studied and produced on an industrial scale for commercial use in the biofuels industry.

One of the ways with which enzymes interact with and bind their substrates into the active site is by charge groups on one molecule, i.e. the enzyme, attracted to the oppositely charged groups on the other molecule, i.e. the substrate. Consequently, if the pH of the environment in which the enzyme-substrate interaction takes place changes, it is possible that the positively and negatively charged groups on the enzyme and/or the substrate change or even lose their charge due to protonization or deprotonization events. The net result of these pH-induced changes is that the enzyme and its substrate will no longer interact with each other in an optimized fashion.

Similar to enzymes optimized to work best at either high or low temperatures, different enzymes are optimized to work at different pH values. For example, enzymes that are present in the acid environment of an animal's stomach are optimized to work at low acidic pH values around pH 3, while pancreatic enzymes that are secreted into the lumen of the small intestine have their highest enzyme activity at neutral or basic pH values (pH 7.2-9.0). Certain bacteria which thrive in alkaline lakes produce enzymes which have the pH optimum of their enzymatic activity at very alkaline pH values.

The objective of this experiment is to study whether and how the enzymatic activity of the cellobiase enzyme changes at different pH values. Instead of using cellobiose as enzyme substrate, an artificial sugar substrate p-nitrophenyl glucopyranoside is used. On catalysis by cellobiase enzyme, the substrate is broken down into glucose, and the yellow substance p-nitrophenol is produced, which absorbs visible light and can be quantitatively measured using a spectrophotometer at a wavelength of 410nm. The solution of interest is placed in a cuvette, which is essentially a test tube with flat walls. A beam of light is transmitted through the solution, and the absorbance (A) measured by the spectrophotometer is directly proportional to the concentration (c) of coloured solution within the cuvette, i.e. $A \propto c$.

Materials and Methods

Experimental setup

Four plastic cuvettes of 1 mL were labelled with numbers corresponding to different pH levels: 1 for pH 3.2, 2 for pH 5.0, 3 for pH 6.3, and 4 for pH 8.6. Each cuvette received 0.5 mL of "Stop" solution using a micropipette.

Four microcentrifuge tubes were also labelled with the same pH values. Next, 0.25 mL of a 3.0 mM "Substrate" solution was added to each tube, followed by 0.25 mL of buffer solution corresponding to the labelled pH value. The contents were mixed well using a vortex for 20 seconds.

Reaction procedure and sample analysis

To initiate the enzyme reaction, 0.25 mL of "Enzyme" solution was quickly added to each microcentrifuge tube, and a stopwatch was started. The tubes, along with a blank sample without the enzyme solution, were then incubated for 5 minutes at room temperature.

After incubation, 0.5 mL of each enzymatic reaction and the blank sample were transferred to the corresponding cuvettes labelled with the appropriate pH and containing the "Stop" solution.

Using a spectrophotometer, the absorbance of the blank sample was measured at 410 nm. This measurement served as the baseline and was used to correct the absorbance values for p-Nitrophenol at 410 nm.

Data recording

The absorbance values for the cuvettes containing the substrate, along with the corrected absorbance values for the blank sample, were then recorded. These values were used to calculate the amount of p-Nitrophenol product (in nmol/mL) for each enzymatic reaction at different pH levels.

The initial rate of each enzymatic reaction was calculated, assuming no p-Nitrophenol was present at the start of the experiment. Finally, a standard curve illustrating the effect of pH on cellobiase enzyme activity was plotted.

Results

The absorbance of each sample was measured at a wavelength of 410 nm and the values were recorded (see Table 1).

Table 1: Absorbance values for known amount of p-nitrophenol.

Amount of p-Nitrophenol (nmol/mL)	Absorbance at 410nm
0	0
12	0.22
24	0.42
48	0.82
72	1.21
96	1.64

A standard curve was produced using the absorption values of the different amounts of p-nitrophenol (see Figure 1 below). This was used to determine the amount of p-nitrophenol product produced by the cellobiase enzyme under different experimental (pH) conditions.

Using the data in table 1, a standard curve with the individual amounts of p-nitrophenol (x-axis) against the corresponding absorbance (y- axis) was plotted and the "best line" connecting all the data points are shown. The equation for the standard curve was determined $y = 0.0163x$ [equation of a straight line $(y) = mx + c$, m is the slope of the line and c is the value where the line cuts the y-axis] and used later to determine the amount of p-nitrophenol product present at different pH.

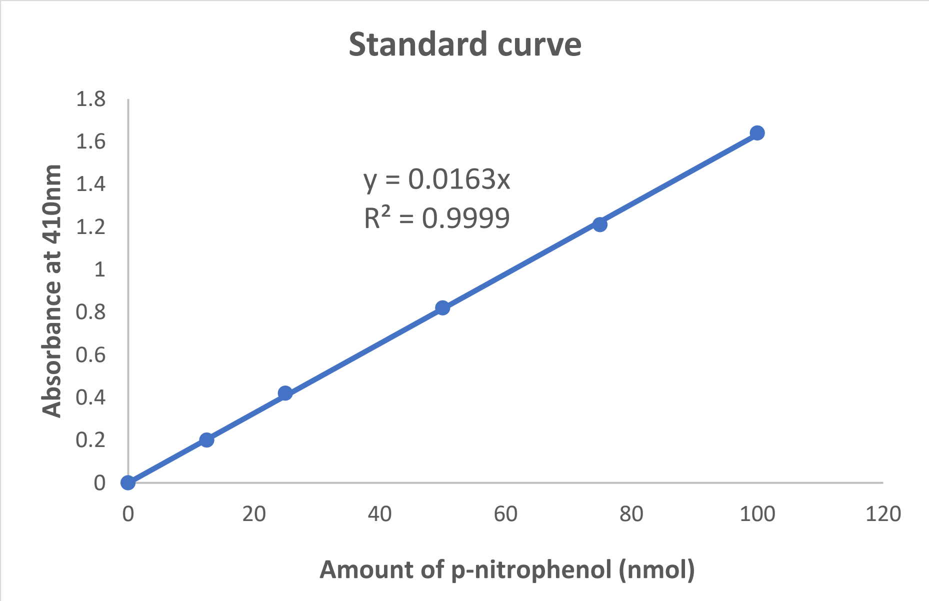

Figure 1: Standard curve for p-Nitrophenol showing the equation y = 0.0163x and R^2= 0.99

The absorbance data for each of the enzyme catalysed reaction at different pH is recorded in Table 2. The absorbance value ranges from 0.12 to 0.84 with highest value at pH 5.0. Using this data together with the earlier established standard curve, the amount of p-Nitrophenol at different pH was calculated.

For e.g. at pH 5.0,

Standard equation , **y = 0.0163x** where y corresponds to the absorbance values and x corresponds to the amount values

Therefore by rearranging the standard equation to obatin the unknown amount, we get,

x = y / 0.0163

x = 0.84 / 0.0163

x = 51.53 nmol/mL

At pH 5.0, the reaction was most optimum with 51.54 nmol of product formed. In very acidic environment (pH 3.2) and alkaline condition (pH 8.6), cellobiases failed to interact well with the substrate and thus very little product (9.20 nmol and 7.36 nmol, respectively) was produced respectively. Since the amount of p-Nitrophenol was measured at one time point (=5min), assume that the amount p-Nitrophenol at 0 minutes was 0 nmol. The initial rate of product formation was calculated as below:

E.g. at pH 3.2

Initial rate of p-Nitrophenol formation = (change in substrate used) / (change in time)

Initial rate of p-Nitrophenol formation = (9.20 nmol – 0 nmol) / (5 min – 0 min)

= 1.84 nmol/min

The data shows that at pH 5.0, the enzyme activity (nmol/min) was maximum and most optimised with 10.30 nmol produced per minute (see Figure 2 below).

Table 2: Absorbance readings of the enzyme reactions and amount of p-nitrophenol determined from the standard curve. The initial rate of product formed by the cellobiase enzyme in the four different pH condition (nmol/min)

pH	Absorbance at 410nm	Amount of p-Nitrophenol (nmol/mL)	Initial rate of p-Nitrophenol formation (nmol/min)
Blank	0	0	0
pH 3.2	0.15	9.20	1.84
pH 5.0	0.84	51.54	10.30
pH 6.3	0.34	20.85	4.17
pH 8.6	0.12	7.36	1.47

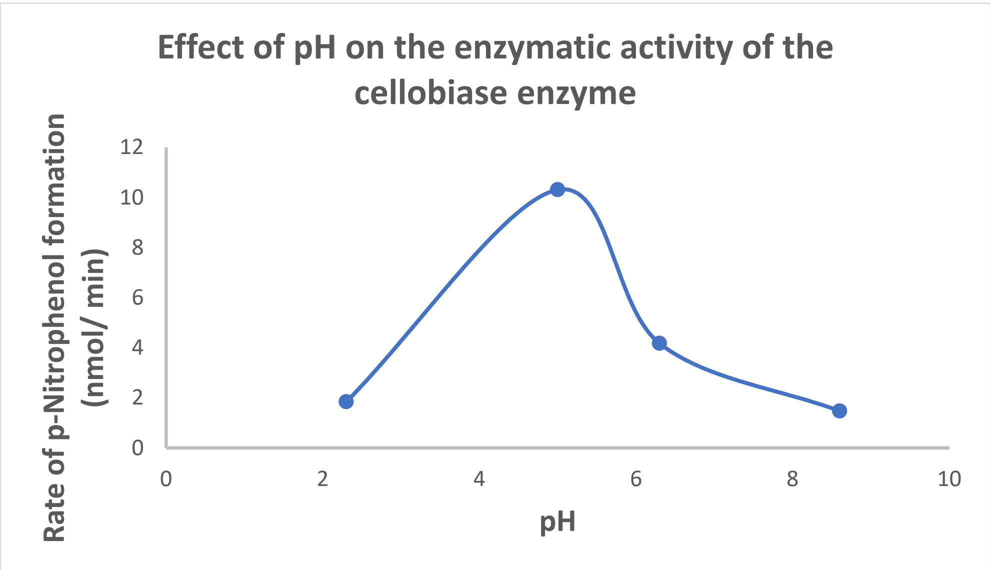

Figure 2: Reaction rate curve for cellobiase enzyme at different pH values. The rate of product formed per minute (y-axis) against pH values (x-axis)

Discussion

Catalases such as cellobiases are important members of the biomass-degrading enzyme system. It hydrolyzes the glucosidic bond of cellobiose and produces two molecules of fermentable glucose which is further used for biofuels production such as ethanol, and butanol. Cellobiases with high activity and great stability over high temperature and varied pH are always preferred in industrial practice for large-scale biomass production. Cellobiases sensitivity to pH is of particular interest because some methods for biomass pre-treatment involves acidic or alkaline condition (Mosier et al 2005). This leads to considerations regarding strategies for neutralization and potential pH drift and assessment of these questions requires knowledge of how sensitive the enzymatic process is to changes in pH.

By utilising an artificial sugar substrate p-nitrophenyl glucopyranoside, the enzyme activity of cellobiase to cleave glucose into the yellow substance p-nitrophenol at different pH was quantitatively measured using spectrophotometer. Based on the reaction rate curve at each pH, the data suggests that for the reaction between the substrate and cellobiase to produce increased amounts of p- Nitrophenol; a pH of 5.0 would be the most preferrable, as it held the highest reaction rate of 10.30 nmol of p- Nitrophenol produced in a minute. Many works have reported traditional bell-shaped pH profiles with an apex (i.e. optimal pH value) around 4-5.

The result of the experiment supports the hypothesis that the rate of enzyme activity will reduce when the pH is below or above the optimum pH. All enzymes have an optimum pH at which they function best. Enzymes also have a working range of pH values at which they can still function well. The active site is formed by a specific conformation of the enzyme's amino acid side chains, which has a specific ionization state that forms the resulting specific three-dimensional structure of the enzyme. Changes in pH affects the ionization state of the amino acids which affects the ionic bonds that hold the three-dimensional shape of the enzyme, thereby impacting the effectiveness of the enzyme activity. In living systems, biochemical buffers maintain an optimum pH range for the enzyme to function best. Extreme pH conditions will cause enzyme to denature. Denaturation is a permanent and irreversible change caused by the breaking of the hydrogen and ionic bonds that maintain the three-dimensional shape of the enzyme. Similar to high temperature, acidic and basic pH can disrupt the three-dimensional shape of the enzyme and change the shape of the active site and therefore causing an irreversible change in the function of the enzyme. When the enzyme becomes denatured and loses its structure, the substrate will not bind to the denatured active site. On the other hand, pH also affects the charge and the shape of the substrate, therefore the substrate cannot bind to the active site.

The limitation of this study is only one measurement per sample is conducted. Future experiment can be improved by increasing sample size and types to repeat more trials to obtain average values of the replicates, therefore reducing random errors. Furthermore, the experiment design can be improved by investigating more pH conditions to obtain a more accurate data to determine the optimum pH.

Conclusion

Cellobiase, an important biomass-degrading enzyme hydrolyzes the glucosidic bond of cellobiose and produces two molecules of fermentable glucose which can be further used for biofuels production. By utilising an artificial sugar substrate p-nitrophenyl glucopyranoside, the enzyme activity of cellobiase to cleave glucose into the yellow substance p-nitrophenol at different pH was quantitatively measured using spectrophotometer at 410nm. Based on the reaction rate curve at each pH, the data suggests that for the reaction between the substrate and cellobiase to produce increased amounts of p- Nitrophenol; a pH of 5.0 would be the most preferrable, as it held the highest reaction rate of 10.30nmol of p- Nitrophenol produced in a minute. Cellobiases activity at various pH is of particular interest because the biofuel production process involves the biomass pre-treatment which is either at acidic or alkaline pH condition (Mosier et al 2005). These findings are important as cellobiases with high activity and great stability over high temperature and varied pH are always preferred in industrial practice for large-scale biofuels production.

References

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